

Bellabeat Case Study: Smart Device Usage Trends

Using Fitbit user data to inform Bellabeat marketing strategy

Executive Summary

This report analyzes Fitbit fitness tracker data to identify smart device usage trends that can inform Bellabeat marketing strategy. The analysis focused on activity, sleep, and weight tracking behavior, with the strongest insights coming from daily activity and sleep records.

Business Task

Analyze smart device usage data to identify consumer behavior trends and provide high-level marketing recommendations for Bellabeat products and the Bellabeat app.

Dataset and Cleaning Summary

Dataset	Records	Unique users	Notes
Daily Activity	1,397	35	Primary dataset; no duplicate rows
Sleep Day	410 after cleaning	24	3 duplicate rows removed
Weight Log	98 after cleaning	13	2 duplicate rows removed; Fat column removed

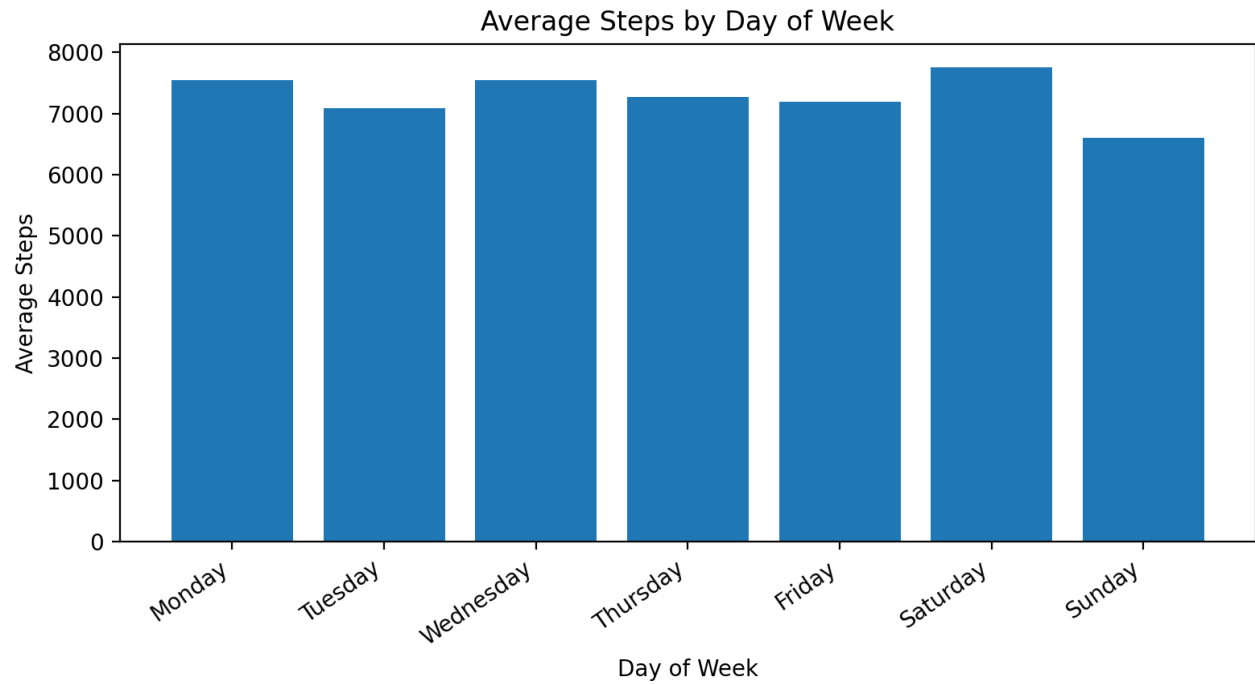
Key Findings

- **Moderate daily activity:** Users averaged 7,281 steps per day, below the commonly referenced 10,000-step benchmark.
- **High sedentary behavior:** Users averaged 993 sedentary minutes per day, or about 16.5 hours.
- **Light activity dominates:** Users averaged 185 minutes of light activity, compared with 20 very active minutes and 13 fairly active minutes per day.
- **Saturday was most active:** Saturday had the highest average step count at 7,752 steps; Sunday had the lowest at 6,607 steps.
- **Sleep varied by weekday:** Users slept most on Sunday at 7.55 hours and least on Thursday at 6.69 hours.

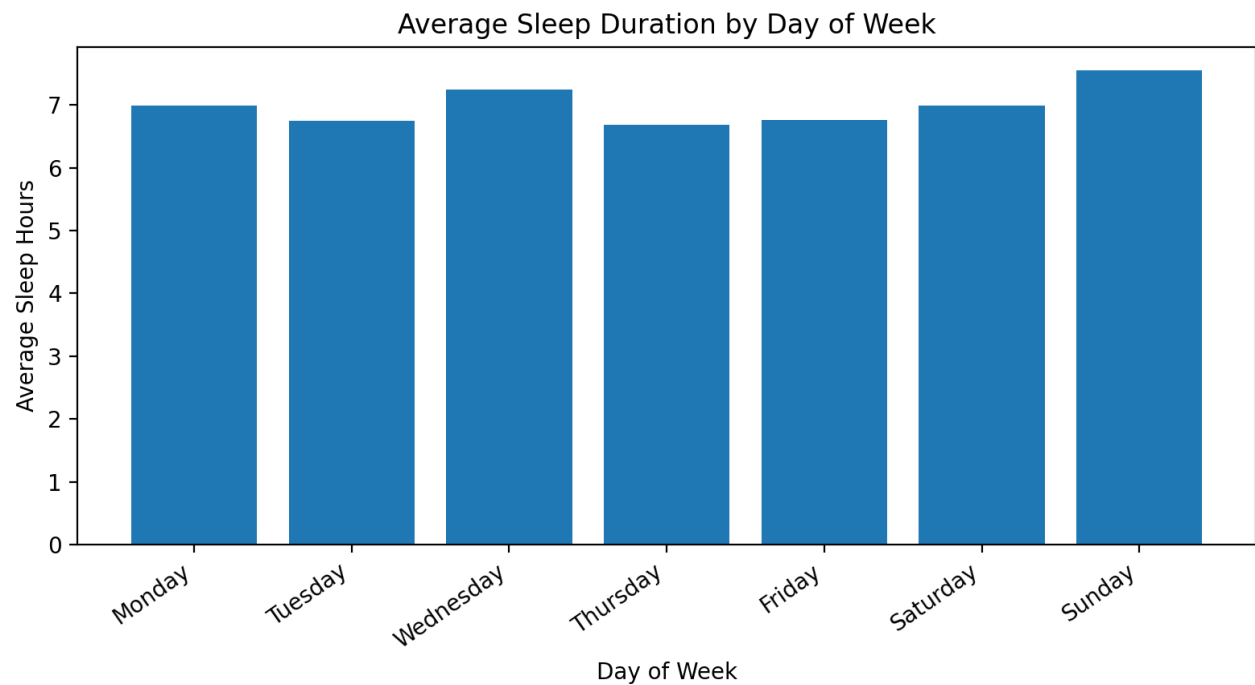
Marketing Meaning

The findings suggest Bellabeat should emphasize movement reminders, step challenges, sustainable habit-building, and weekly sleep insights. Activity tracking appears to be the strongest entry point for user engagement, while sleep insights can deepen the wellness message.

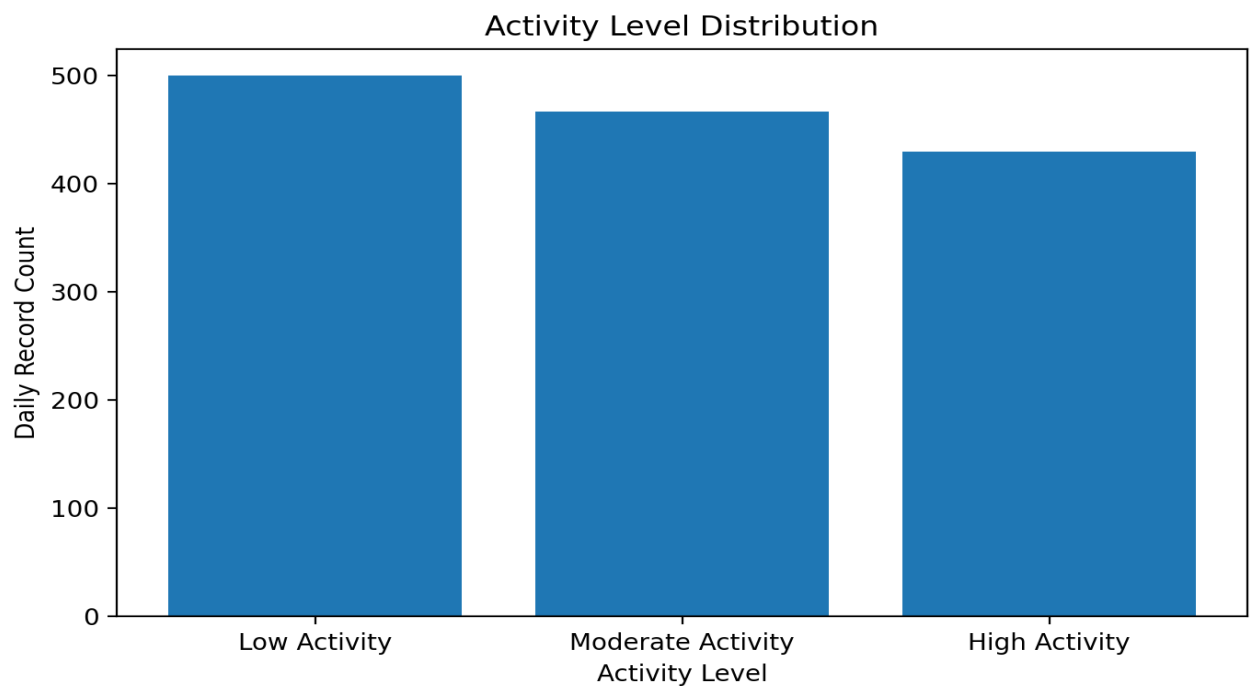
Supporting Visualizations



Average steps by day of week: Saturday was highest and Sunday was lowest.



Average sleep by day of week: Sunday was highest and Thursday was lowest.



Activity level distribution: Low Activity was the largest category.

Key Metrics Summary

Metric	Value
Average daily steps	7,281
Average sleep	7.0 hours
Average sedentary time	16.5 hours
Activity users	35
Sleep users	24
Weight users	13

Recommendations

1. Promote movement reminders and step challenges

Because sedentary time was high and users averaged below 10,000 steps per day, Bellabeat should emphasize hourly movement reminders, short walking goals, and weekly step challenges.

2. Market Bellabeat as a holistic wellness companion

Since users engage with both activity and sleep tracking, Bellabeat should position its products around overall wellness, connecting movement, rest, mindfulness, and daily routines.

3. Use personalized weekly insights to improve engagement

Bellabeat can increase user engagement through weekly summaries, habit streaks, sleep insights, and personalized coaching prompts inside the Bellabeat app.

Limitations

The dataset has several limitations: small sample size, 2016 collection period, no demographic information, limited weight-tracking participation, and Fitbit users rather than Bellabeat customers directly. Findings should be treated as directional insights rather than definitive conclusions.

Conclusion

The analysis suggests that smart device users engage most consistently with activity tracking, spend substantial time sedentary, and show meaningful weekly patterns in movement and sleep. Bellabeat can use these insights to strengthen marketing around personalized wellness, movement encouragement, sleep support, and habit-building features.